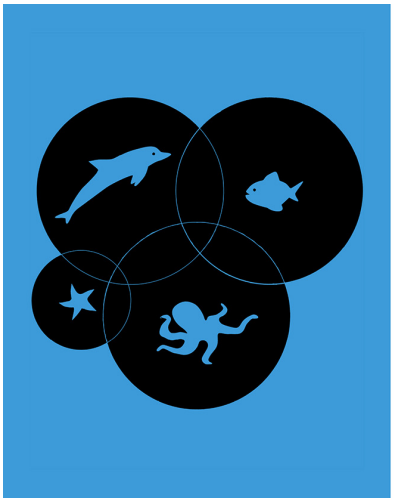


EXISTENCE IS DETERMINED BY A SLENDER THREAD OF CIRCUMSTANCES

ECOLOGICAL NICHES

IN CONTEXT

KEY FIGURE

Joseph Grinnell (1877–1939)

BEFORE

1910 In a paper about beetles, Roswell Hill Johnson, a US biologist, is the first person to use the word “niche” in a biological context.

AFTER

1927 British ecologist Charles Elton stresses the importance of an organism’s role as well as its “address” in his definition of an ecological niche in his book *Animal Ecology*.

1957 In an academic paper called “Concluding Remarks,” British ecologist George Evelyn Hutchinson expands the theory of niches to embrace an organism’s entire environment.

1968 A study by Australian D.R. Klein of the introduction, increase, and die-off of reindeer on St. Matthew Island, Alaska, identifies the destructive niche.

An organism’s niche is a combination of its place and its role in the environment. It encompasses how the organism meets its needs for food and shelter, as well as how it avoids predators, competes with other species, and